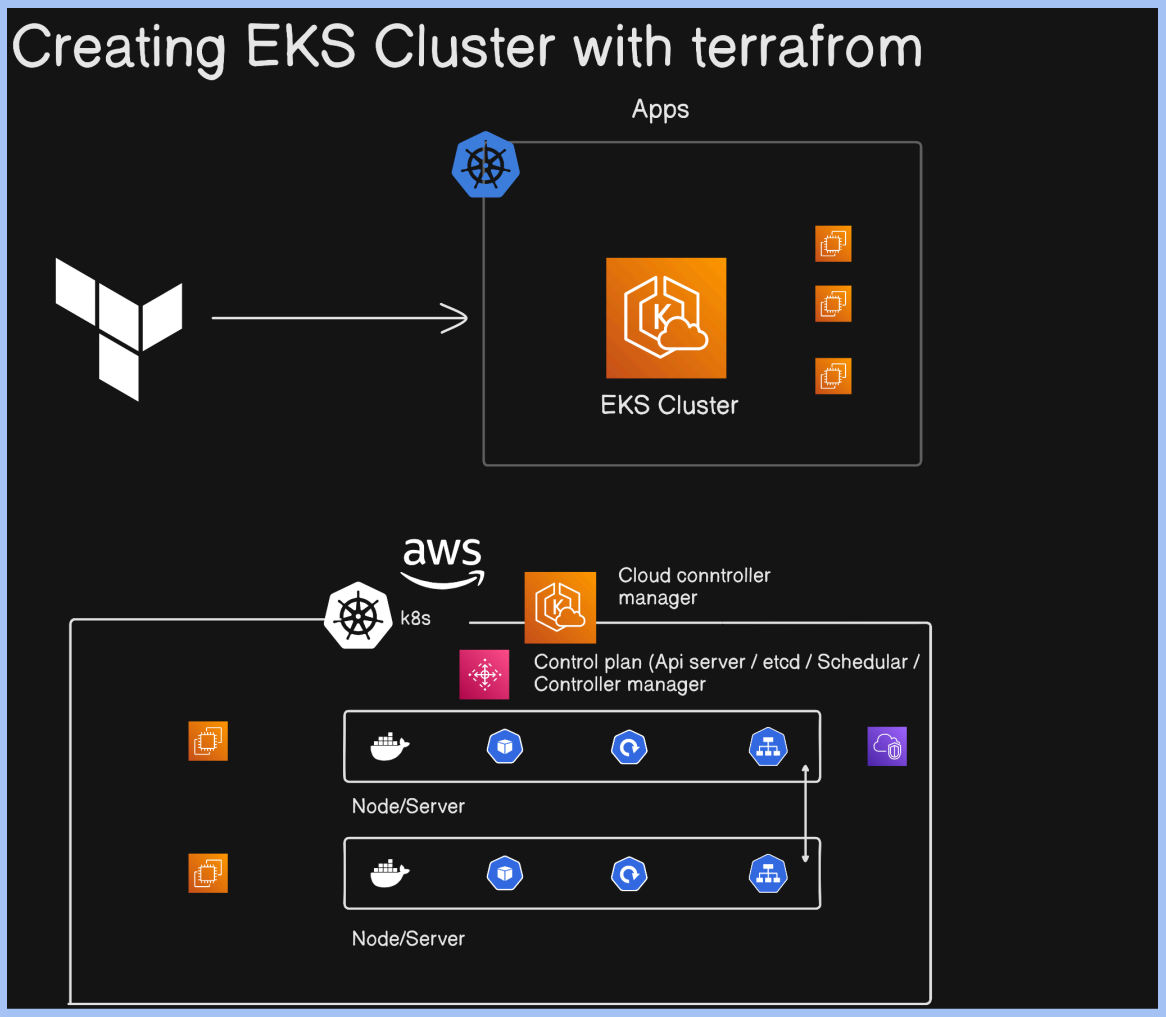
Terraform AWS EKS Cluster Setup



This project provisions an **Amazon Elastic Kubernetes Service (EKS) cluster** using **Terraform** and the official AWS EKS module. The configuration follows best practices, including networking, security, and cost-effective node groups.

🚀 Features

- **EKS Cluster (v1.29)** with public access enabled
- **VPC Configuration** with private, public, and intra subnets
- **Managed Node Group** with a **SPOT + ON_DEMAND** instance strategy
- **IAM Role for Node Groups** to ensure proper permissions
- **Essential Cluster Add-ons** (VPC CNI, CoreDNS, Kube Proxy)

- **Security Features** (Encryption for secrets)

📌 Prerequisites

Ensure you have the following installed:

- Terraform (>=1.3.0)
- AWS CLI configured with appropriate permissions
- IAM permissions to create VPCs, EKS clusters, and IAM roles

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📂 Folder Structure

```
├── main.tf                # Main Terraform configuration
├── vpc.tf                 # VPC configuration
├── eks.tf                 # EKS Cluster setup
├── variables.tf           # Variables for Terraform modules
├── outputs.tf             # Outputs for Terraform modules
└── README.md              # Project Documentation
```

⚙️ Configuration

1️⃣ Terraform Provider

```
terraform {
  required_version = ">= 1.3.0"
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.92.0"
    }
  }
}
```

2️⃣ Local Variables

```
locals {
  region         = "us-east-2"
  name           = "tes-dev-eks-cluster"
  vpc_cidr      = "10.0.0.0/16"
  azs           = ["us-east-2a", "us-east-2b"]
  private_subnets = ["10.0.1.0/24", "10.0.2.0/24"]
  public_subnets  = ["10.0.101.0/24", "10.0.102.0/24"]
  intra_subnets   = ["10.0.5.0/24", "10.0.6.0/24"]
  env            = "dev"
}
```

3️⃣ VPC Module

```
module "vpc" {
  source  = "terraform-aws-modules/vpc/aws"
  name    = "${local.name}-vpc"
  cidr    = local.vpc_cidr
  azs             = local.azs
  private_subnets = local.private_subnets
  public_subnets  = local.public_subnets
  intra_subnets   = local.intra_subnets
  enable_nat_gateway = true
  enable_vpn_gateway = true
  tags = {
    Terraform   = "true"
    Environment = local.env
  }
}
```

4️⃣ EKS Module

```
module "eks" {
  source  = "terraform-aws-modules/eks/aws"
  version = "~> 20.31"
  cluster_name                   = local.name
  cluster_version                = "1.29"
  cluster_endpoint_public_access = true
  vpc_id     = module.vpc.vpc_id
  subnet_ids = module.vpc.private_subnets
  cluster_addons = {
    vpc-cni = {
      most_recent = true
      addon_version = "v1.14.1-eksbuild.1"
    }
    kube-proxy = {
      most_recent = true
    }
    coredns = {
      most_recent = true
      addon_version = "v1.10.1-eksbuild.1"
    }
  }
  eks_managed_node_group_defaults = {
    instance_types = ["t2.micro"]
    attach_cluster_primary_security_group = true
    iam_role_name = "eks-node-group-role"
  }
  eks_managed_node_groups = {
    tws-cluster-ng = {
      instance_types = ["t2.micro"]
      min_size     = 2
      max_size     = 3
      desired_size = 2
      capacity_type  = "SPOT"
      on_demand_percentage_above_base_capacity = 50
    }
  }
  tags = {
    Environment = local.env
    Terraform   = "true"
  }
}
```

🛠️ Deployment Steps

1️⃣ Initialize Terraform

```
terraform init
```

2️⃣ Plan Deployment

```
terraform plan
```

3️⃣ Apply Changes

```
terraform apply -auto-approve
```

4️⃣ Destroy Resources (If Needed)

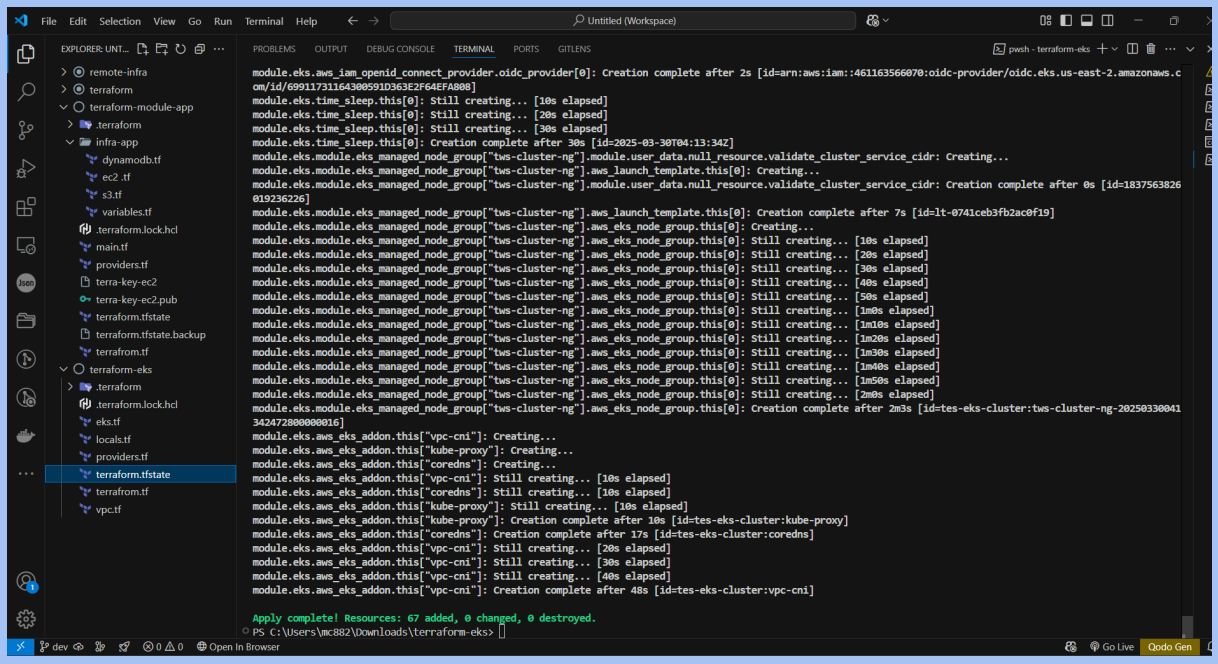
```
terraform destroy -auto-approve
```

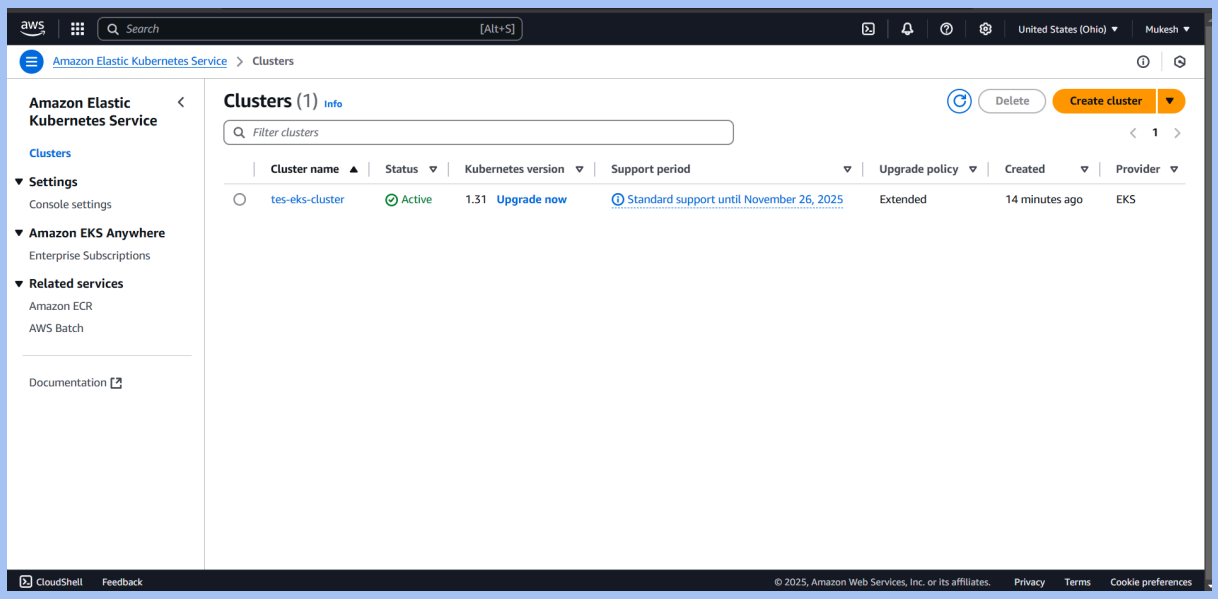
📌 Best Practices Implemented

✅ Used **SPOT + ON_DEMAND** node group strategy for cost efficiency.
✅ IAM role added for **EKS Managed Node Group**.
✅ Latest stable **EKS version (1.29)** used.
✅ Explicit **addon versions** to avoid compatibility issues.
✅ Enabled **encryption for secrets** using AWS KMS.
✅ NAT Gateway enabled for private subnet connectivity.

🎯 Next Steps

- Implement **IAM Role for Service Accounts (IRSA)** for better security.
- Integrate **CloudWatch/Prometheus** for monitoring.
- Set up **GitHub Actions CI/CD** for EKS deployment automation.


File Edit Selection View Go Run Terminal Help
Untitled (Workspace)
EXPLORER: UNT...
remote-infra
terraform
terraform-module-app
.terraform
infra-app
dynamodb.tf
ec2 .tf
s3.tf
variables.tf
.terraform.lock.hcl
main.tf
providers.tf
terra-key-ec2
terra-key-ec2.pub
terraform.tfstate
terraform.tfstate.backup
terrafrom.tf
terraform-eks
.terraform
.terraform.lock.hcl
eks.tf
locals.tf
providers.tf
terraform.tfstate
terrafrom.tf
vpc.tf
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
pwsh - terraform-eks
module.eks.aws_iam_openid_connect_provider.oidc_provider[0]: Creation complete after 2s [id=arn:aws:iam::461163566070:oidc-provider/oidc.eks.us-east-2.amazonaws.com/id/699117311643005 91D363E2F64EFA808]
module.eks.time_sleep.this[0]: Still creating... [10s elapsed]
module.eks.time_sleep.this[0]: Still creating... [20s elapsed]
module.eks.time_sleep.this[0]: Still creating... [30s elapsed]
module.eks.time_sleep.this[0]: Creation complete after 30s [id=2025-03-30T04:13:34Z]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].module.user_data.null_resource.validate_cluster_service_cidr: Creating...
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_launch_template.this[0]: Creating...
module.eks.module.eks_managed_node_group["tws-cluster-ng"].module.user_data.null_resource.validate_cluster_service_cidr: Creation complete after 0s [id=1837563826019236226]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_launch_template.this[0]: Creation complete after 7s [id=lt-0741ceb3fb2ac0f19]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Creating...
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [10s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [20s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [30s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [40s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [50s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [1m0s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [1m10s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [1m20s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [1m30s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [1m40s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [1m50s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Still creating... [2m0s elapsed]
module.eks.module.eks_managed_node_group["tws-cluster-ng"].aws_eks_node_group.this[0]: Creation complete after 2m3s [id=tes-eks-cluster:tws-cluster-ng-20250330041342472800000016]
module.eks.aws_eks_addon.this["vpc-cni"]: Creating...
module.eks.aws_eks_addon.this["kube-proxy"]: Creating...
module.eks.aws_eks_addon.this["coredns"]: Creating...
module.eks.aws_eks_addon.this["vpc-cni"]: Still creating... [10s elapsed]
module.eks.aws_eks_addon.this["coredns"]: Still creating... [10s elapsed]
module.eks.aws_eks_addon.this["kube-proxy"]: Still creating... [10s elapsed]
module.eks.aws_eks_addon.this["kube-proxy"]: Creation complete after 10s [id=tes-eks-cluster:kube-proxy]
module.eks.aws_eks_addon.this["coredns"]: Creation complete after 17s [id=tes-eks-cluster:coredns]
module.eks.aws_eks_addon.this["vpc-cni"]: Still creating... [20s elapsed]
module.eks.aws_eks_addon.this["vpc-cni"]: Still creating... [30s elapsed]
module.eks.aws_eks_addon.this["vpc-cni"]: Still creating... [40s elapsed]
module.eks.aws_eks_addon.this["vpc-cni"]: Creation complete after 48s [id=tes-eks-cluster:vpc-cni]
Apply complete! Resources: 67 added, 0 changed, 0 destroyed.
PS C:\Users\mc882\Downloads\terraform-eks>
dev
0 0
Open In Browser
Go Live
Qodo Gen


aws
Search
[Alt+S]
United States (Ohio)
Mukesh
Amazon Elastic Kubernetes Service > Clusters
Amazon Elastic Kubernetes Service
Clusters
Settings
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Related services
Amazon ECR
AWS Batch
Documentation
Clusters (1) Info
Delete
Create cluster
Filter clusters
1
Cluster name
Status
Kubernetes version
Support period
Upgrade policy
Created
Provider
tes-eks-cluster
Active
1.31 Upgrade now
Standard support until November 26, 2025
Extended
14 minutes ago
EKS
CloudShell
Feedback
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Amazon Elastic Kubernetes Service > Clusters > tes-eks-cluster

Amazon Elastic Kubernetes Service

- Clusters
- Settings
  - Console settings
- Amazon EKS Anywhere
  - Enterprise Subscriptions
- Related services
  - Amazon ECR
  - AWS Batch

Documentation

tes-eks-cluster

Delete cluster | Upgrade version | View dashboard

Your current IAM principal doesn't have access to Kubernetes objects on this cluster.
This may be due to the current user or role not having Kubernetes RBAC permissions to describe cluster resources or not having an entry in the cluster's auth config map. Learn more

End of standard support for Kubernetes version 1.31 is November 26, 2025. On that date, your cluster will enter the extended support period with additional fees. For more information, see the pricing page.

Upgrade now

Cluster info Info

Status	Kubernetes version Info	Support period	Provider
Active	1.31	Standard support until November 26, 2025	EKS

Cluster health issues	Upgrade insights	Node health issues
0	0	0

Overview | Resources | Compute | Networking | Add-ons 1 | Access | Observability | Update history | Tags

Details

API server endpoint	OpenID Connect provider URL	Created

CloudShell | Feedback

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